

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – TECHNICAL PRESENTATION

Course Code:	CSA16 10	Course Title:	Data Warehousing and Data Mining
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Technical Presentation
Weightage:	10% of CIA	Total Marks:	100 (2 Question × 100 Marks)
CO1 Statement:	<i>Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2</i>		
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Section / Batch:		Date:	20/07/2026

Instructions to Students

- Answer the given question through a technical presentation. Total Marks: 100.
- Clearly define the problem and explain the concept of algorithm growth functions.
- Present comparative analysis using appropriate representations such as graphs, tables, or charts.
- Use suitable tools (PowerPoint, visualization tools, or software) to support your explanation.
- Ensure technical accuracy, logical flow, and proper interpretation of results.
- Maintain clarity in communication, proper slide organization, and professional delivery.
- Time management, confidence, and audience engagement will be considered during evaluation.

Question Type	Focus Area	Questions	Marks
Technical Presentation	Decision support system, Operational versus Decision support System, Architecture of data warehouse, ETL, OLAP and data cubes, Dimensional data modelling -star snowflake schemas. Query Processing, Open-source Business Intelligence Tools. Develop an application to perform OLAP operations - Introduction to KNIME tool for reporting.	Q1, Q2	100
GRAND TOTAL:			100 Marks

SECTION A – Technical Presentation

Topic 1: Decision Support Systems (DSS)

☉ Focus Areas:

- Definition and need of DSS
- Components (Data, Model, UI)
- Types of DSS
- Real-world applications (banking, healthcare)
- Advantages & limitations

Topic 2: Operational Systems vs DSS

☉ Focus Areas:

- OLTP vs OLAP concepts
- Key differences (data, users, purpose)
- Performance comparison
- Use-case scenarios
- Integration between systems



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A **Data Warehouse** is a centralized repository that stores integrated, historical, and organized data collected from multiple sources. It is mainly used for reporting, data analysis, and business decision-making. The architecture of a data warehouse defines how data is collected, processed, stored, and accessed by users. A typical data warehouse architecture consists of several important components that work together to provide accurate and reliable information.

The first component is the **Data Sources Layer**, where data is collected from different internal and external sources. Internal sources include operational databases such as sales, finance, human resources (HR), Customer Relationship Management (CRM), and Enterprise Resource Planning (ERP) systems. External sources may include websites, cloud applications, social media, and third-party data providers. Since the data comes in different formats, it must be processed before being stored in the warehouse.

The next component is the **ETL (Extract, Transform, Load) Process**. This is one of the most important stages in data warehousing. During the **Extract** phase, data is collected from various source systems. In the **Transform** phase, the extracted data is cleaned, validated, standardized, and converted into a common format by removing duplicate records and correcting errors. Finally, in the **Load** phase, the processed data is stored in the data warehouse. This ensures that only accurate and high-quality data is available for analysis.

After processing, the data is stored in the **Data Warehouse**, which acts as the central repository. It stores large volumes of historical and subject-oriented data that is optimized for querying and analysis rather than daily transactions. The data is usually organized using schemas such as **Star Schema** or **Snowflake Schema**, which improve query performance and make data retrieval faster.

Many organizations also create **Data Marts**, which are smaller sections of the data warehouse designed for specific departments such as sales, marketing, finance, or human resources. Data marts allow different departments to access only the information they need, improving efficiency and reducing query time.

The **OLAP (Online Analytical Processing)** layer is used to perform multidimensional analysis on the stored data. OLAP enables users to summarize, filter, compare, and analyze information quickly using operations such as roll-up, drill-down, slice, and dice. This helps managers and analysts identify business trends and make informed decisions.

Finally, users access the data warehouse through **Front-End Tools** such as reporting software, dashboards, business intelligence (BI) tools, and data visualization applications. These tools help users generate reports, monitor business performance, and gain valuable insights from the stored data.

In conclusion, the architecture of a data warehouse provides a structured approach for collecting, processing, storing, and analyzing data. Its major components—data sources, ETL process, data warehouse, data marts, OLAP, and front-end tools—work together to support efficient business intelligence and decision-making. A well-designed data warehouse improves data quality, enhances reporting, and enables organizations to make faster and more informed decisions.

